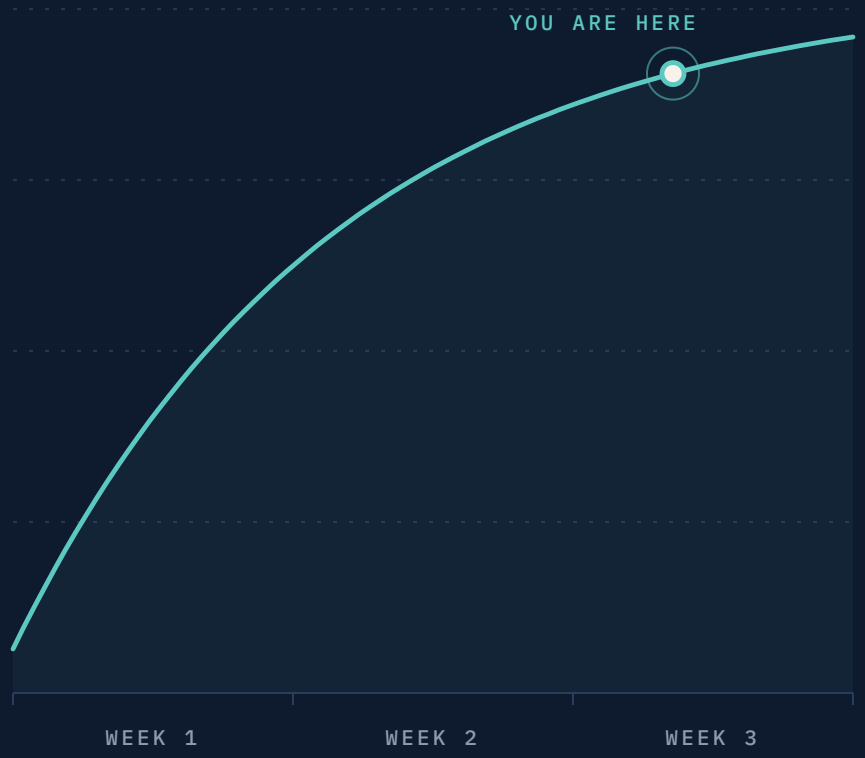
WEEK 3 · DAY 2 · OVERVIEW

Spread Over the Curve



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

Our month-end file has a column called oas. Bezane Finance's 2032 bond shows 147. Where does that come from? Rebuild it for all 200 bonds. Then tell me whether our own ten-tenor curve gives the same spreads, and what we should call the number.

ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

Your Route Through Day 2

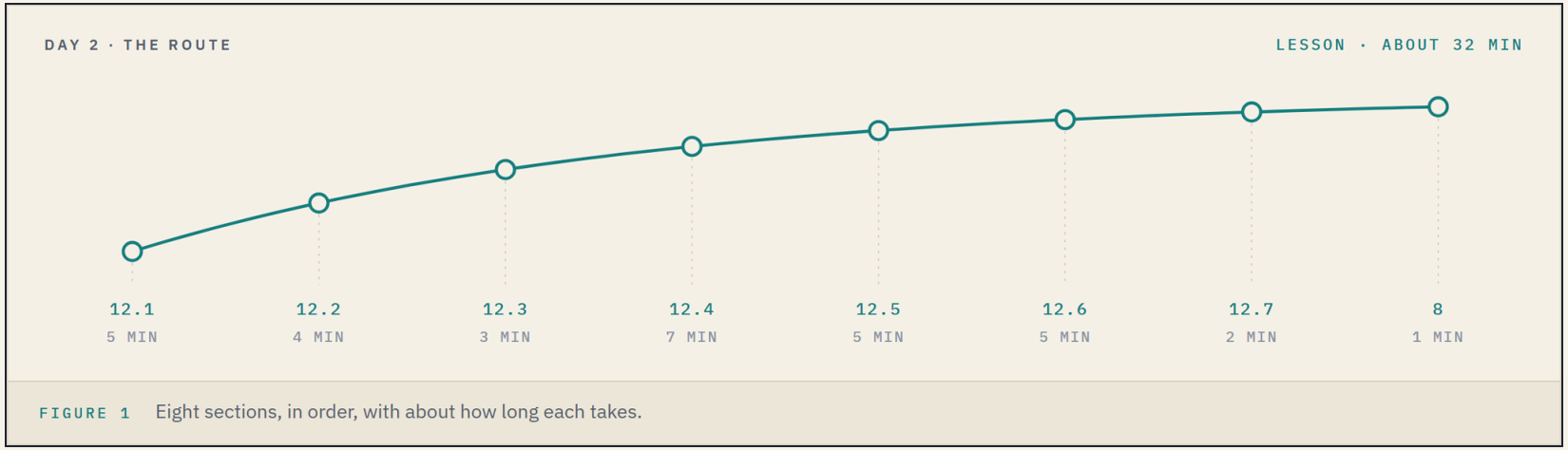


FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
12.1	One Bond, by Hand	5 min
12.2	Years to Maturity, the File's Way	4 min
12.3	The Curve at Bezane's Maturity, and 147	3 min
12.4	Two Functions, Docstring First	7 min

	SECTION	ABOUT
12.5	Tests From Excel, Then Commit	5 min
12.6	The Desk's Ten-Tenor Table, and the Gap	5 min
12.7	What to Call the Number	2 min
8	Bring It Home	1 min

Time. About 32 minutes for the lesson, then about 12 for the four exercises.

12.1 One Bond, by Hand

THE BOOK

	CUSIP	ISSUER	MATURITY	YIELD_PCT	OAS
0	99080CAE8	Bezane Finance	2032-06-18	5.606	147
1	99049XAE2	Bezamyth Machinery	2029-04-17	9.220	535
2	99039NAC0	Bezoquil Freight	2030-06-03	5.509	153
3	99039NAG1	Bezoquil Freight	2053-04-09	6.756	218

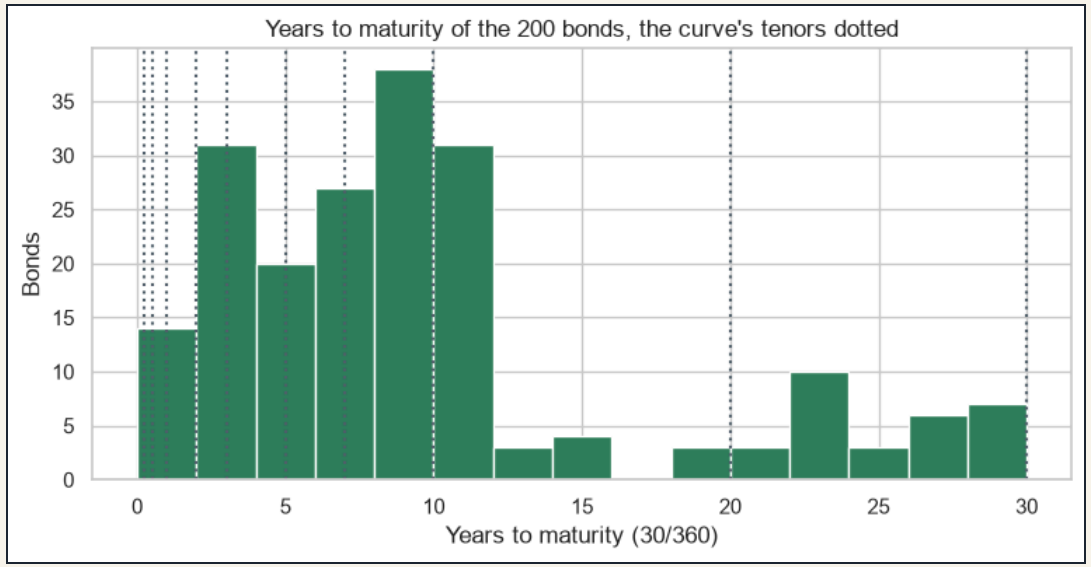
THE CURVE

```
curve = pd.read_csv('course2_treasury_curve.csv')
```

```
[0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0]
[3.587, 3.622, 3.69, 3.811, 3.916, 4.086, 4.214, 4.348, 4.54, 4.586]
```

A spread to the curve is the bond's yield minus the curve's yield at the same maturity.

12.2 Years to Maturity, the File's Way



THE CODE

```
holdings['years'] = [bondmath.days_30_360(a, m) / 360
```

WHAT IT PRINTS

```
1.0056 to 29.8972 years
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Years, 30/360	`=YEARFRAC(D2,E2,0)`	5.716666666666667
The day count	`=YEARFRAC(D2,E2,0)*360`	2058

Every bond sits between two tenors, so each needs the curve read between points.

12.3 The Curve at Bezane's Maturity, and 147

THE FILE'S FORMULA

```
curve_formula_pct = round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)
```

THE SPREAD, FROM THE FORMULA

```
print(f"{(bond['yield_pct'] - curve_formula_pct) * 100:.6f} bp; the file says {bond['oas']}")
```

```
147.000000 bp; the file says 147
```

THE STRAIGHT LINE

```
straight line: 4.131867 percent
```

EXCEL SIDE BY SIDE

JOB	EXCEL
The straight line	notebook 11's `FORECAST.LINEAR` with `INDEX` and `MATCH`
The formula	`=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)`

The file read its curve from the formula at the bond's own years, rounded to 3 decimals: 147 to the last digit.

12.4 Two Functions, Docstring First

WRITE SPREAD_TO_CURVE_BP

```
%%writefile -a bondmath.py
def spread_to_curve_bp(yield_pct: float, years: float, curve_tenors: list[float],
                       curve_yields_pct: list[float]) -> float:
    """Yield spread over a curve at the bond's own maturity, in basis points.
    Units: yield_pct and curve yields in percent, years and tenors in years; the result is in bp.
    """
    curve_yield_pct = interpolate_yield(years, curve_tenors, curve_yields_pct)
    return (yield_pct - curve_yield_pct) * 100
```

THE WHOLE BOOK

```
largest miss: 2.3e-13 bp
```

Two functions rebuild the `oas` column for all 200 bonds, to 2.3e-13 of a basis point.

12.5 Tests From Excel, Then Commit

TEST A CASE WORKED BY HAND

```
%%writefile -a test_bondmath.py
def test_spread_to_curve_known_case():
    curve_at_6_years = 4.086 + (4.214 - 4.086) * (6 - 5) / (7 - 5)
    expected_bp = (5.0 - curve_at_6_years) * 100
    result = bondmath.spread_to_curve_bp(5.0, 6.0, CURVE_TENORS, CURVE_YIELDS_PCT)
    assert result == pytest.approx(expected_bp, abs=1e-9)
```

TEST THE DAY COUNT

```
def test_years_to_maturity_cases():
        (date(2026, 9, 30), date(2036, 9, 30), 3600),  # exactly ten years
        (date(2026, 9, 30), date(2032, 6, 18), 2058),  # the first bond in holdings.csv
        (date(2027, 2, 28), date(2031, 8, 31), 1621),  # the 31st end stays 31 after a 28 February start
        assert bondmath.years_to_maturity(settlement, maturity) == pytest.approx(days / 360, abs=1e-12)
```

35 tests pass, and the history says when every test arrived.

12.6 The Desk's Ten-Tenor Table, and the Gap

EVERY SPREAD ON THE TABLE

```
holdings['curve_table_pct'] = [bondmath.interpolate_yield(t, curve_tenors, curve_yields)
                               for t in holdings['years']]
holdings['spread_table_bp'] = [bondmath.spread_to_curve_bp(y, t, curve_tenors, curve_yields)
                               for y, t in zip(holdings['yield_pct'], holdings['years'])]
holdings['gap_bp'] = holdings['spread_table_bp'] - holdings['oas']
```

```
Bezane Finance on the table: 147.41 bp
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Spread to the table, bp	`=(F2-H2)*100`	147.41333333333327

The same bond on the same curve is 147 or 147.41: compare two spreads only when they share the curve, the reading between tenors, and the count of years.

12.7 What to Call the Number

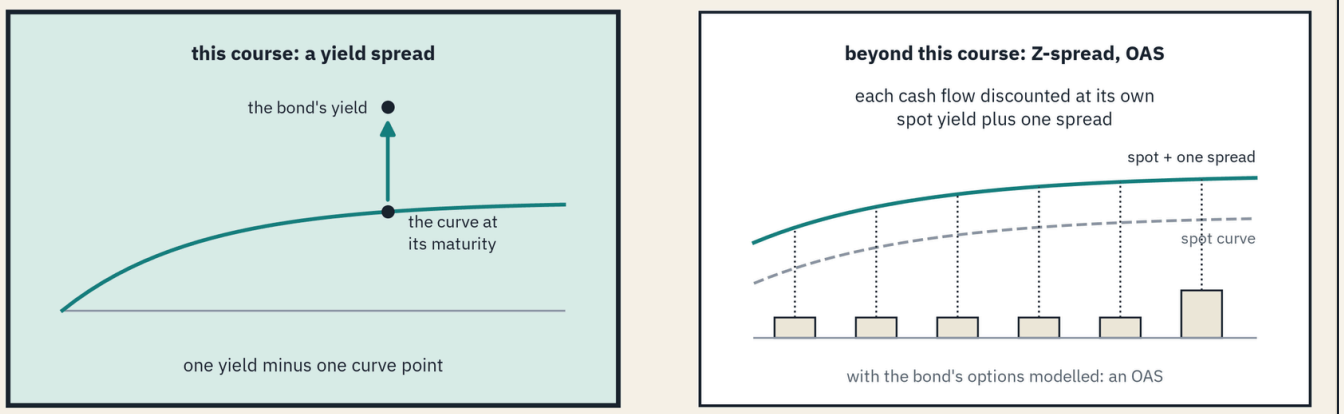


FIGURE 7 A yield spread subtracts one curve point; a Z-spread discounts each cash flow, and an OAS models options.

	OAS	SPREAD_FORMULA_BP	SPREAD_TABLE_BP
0	147	147.0	147.4133
1	535	535.0	535.1542
2	153	153.0	153.5625
3	218	218.0	218.5985
4	127	127.0	127.3713

The course calls it `spread_to_curve_bp`, a yield spread at the bond's own maturity; the file keeps the name `oas`.

Bring It Home

IN POWERSHELL

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_12_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 12: years_to_maturity and spread_to_curve_bp"
git log --oneline -- test_bondmath.py
git diff --stat week2
```

IF WEEK2 IS UNKNOWN

If **week2** is unknown, tag notebook 10's commit: **`git tag week2`** followed by its hash.

IN COLAB

In Colab, download the two files from the work folder in the file browser.

You Can Now

- Say what a spread to the curve measures, in bp.
- Count years to maturity on 30/360 with **`years_to_maturity`**, as Excel's **`YEARFRAC`** with basis 0 does.
- Compute a spread with **`spread_to_curve_bp`** on any curve, a one-point curve included.
- Rebuild the file's **`oas`** column from the curve's formula.
- Measure how far the reading between tenors moves a spread, and say when two spreads compare.
- Call the number a yield spread, and list the commits that changed one file.

NEXT
WEEK 3 · DAY 3

Spread Duration and DTS

Spread duration measured by moving the spread and repricing, DTS built from it, and the book's spread risk by rating bucket.